

JOHN K. RYU

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Full contact information available upon request

PROFESSIONAL EXPERIENCE

AI Engineer | TecAce Software Ltd

Bellevue, WA | April 2025 - Present

- **Architected a governance-first, privacy-first enterprise AI agent platform** with a retrieval-augmented generation (RAG) pipeline (LangChain, LangSmith, LangGraph), deployable on hybrid infrastructure — cloud (AWS) or fully on-prem with self-hosted model serving — so regulated teams keep sensitive data inside their own perimeter
- **Engineered LangGraph-based agentic workflow platform** — visual flow builder with intelligent routing, guardrails, conversation memory, and node-level inference overrides powering multi-agent enterprise applications
- **Built messaging-platform integration gateway** connecting Slack and Microsoft Teams to AI applications with signed webhooks, multi-app alias routing, and real-time SSE streaming replies
- **Developed hybrid search optimization** implementing RRF reranking, cross-encoder reranking, and score fusion techniques for enhanced search relevance across Milvus vector databases
- **Built end-to-end LLM evaluation & quality system** — synthetic test-set generation (RAGAS TestsetGenerator, DeepEval Synthesizer), versioned datasets, and an evaluation runner that replays test cases through the live RAG pipeline, capturing answers, retrieved documents, and citation scores
- **Quantified retrieval and generation quality** with algorithmic metrics (Precision@k, Recall@k, MRR, NDCG, source coverage) plus a configurable LLM-as-judge layer (AWS Bedrock or on-prem vLLM) scoring faithfulness, answer relevance, and hallucination rate — validated against FinanceBench, with A/B run comparison, known-bad-prompt regression suites, and automated drop-vs-baseline detection
- **Delivered external developer platform** — access-key-secured public API gateway plus a complete data-driven API reference documentation portal covering 500+ endpoints across six microservices
- **Championed AI-native engineering practice** orchestrating coding agents (Claude Code) through spec → plan → multi-agent implementation → adversarial QA workflows across a 12-repository platform

e-Commerce Engineer/Owner | Roem Ventures LLC

Seattle, WA | May 2022 - Present

- **Founded and developed e-commerce brand from concept to market launch** including comprehensive product development, logistics coordination, and business operations management
- **Engineered custom Shopify solutions using Liquid programming** developing tailored plugins and functionality enhancements for optimized user experience
- **Implemented integrated marketing campaign strategies** utilizing Google AdWords and Facebook Advertising platforms for targeted customer acquisition and conversion optimization

Software Engineer - Xfinity X1 & Flex | Comcast

Philadelphia, PA | Oct 2017 - Nov 2021

- **Successfully developed and maintained Xfinity Flex Streaming product from ground up** building key middleware APIs that power core Flex features, resulting in positive customer feedback
- **Exceeded subscription acquisition targets by 1000%** surpassing initial goal of 100K subscriptions within six months by acquiring over 1 million subscriptions
- **Developed and maintained core features for cloud-based X1 Guide interface** directly impacting over 31 million customers across Xfinity TV platform
- **Implemented scalable backend architecture** using distributed systems design principles to support millions of concurrent streaming users

Startup Founder | Parkgorithm

Philadelphia, PA | May 2016 - Jun 2017

- **Founded startup tackling urban street parking through mobile technology** — led cross-functional team, product roadmap, and Android MVP delivery

Software Engineer | Graphite GTC

Bryn Mawr, PA | May 2015 - May 2016

- **Developed Android and Web client applications** for the proprietary Graphite platform — responsive, cross-platform UIs with UX and performance optimization

Undergraduate Researcher | Temple University

Philadelphia, PA | Jun 2015 - Oct 2015

- **Conducted computer vision research under NSF funding** developing machine learning algorithms for mobile robot object detection and recognition systems
- **Developed innovative "Supported_By" algorithm in MATLAB** enabling spatial relationship detection for autonomous robotics applications

Founder/Sales Director | Fatty Pocket

Norristown, PA | Sept 2011 - Jul 2013

- **Founded and scaled digital coupon-marketing platform** for local businesses — led business development, sales teams, and strategic vendor relationships

TECHNICAL SKILLS

Agentic / AI-Native Development:

Coding Agents (Claude Code, Codex) • Multi-Agent Orchestration • MCP (Model Context Protocol) • Prompt Engineering • Agent-Assisted SDLC • LangGraph Workflows

Frontend:

React.js • Next.js 14 • TypeScript • JavaScript ES6+ • HTML5/CSS3 • Tailwind CSS • Responsive Design

Databases:

PostgreSQL • MySQL • DynamoDB • MongoDB • Vector Databases (Milvus) • Redis/Valkey

Backend:

Python • FastAPI/Flask • Java • Spring Boot • Node.js/Express • RESTful APIs • SSE Streaming • GraphQL • Microservices • OpenAPI/Swagger

AI/ML:

RAG Pipeline Architecture • Hybrid Search (Sparse + Dense) • Embeddings & Semantic Search • Vector Databases • Cross-encoder Reranking • LLM Evaluation (RAGAS, DeepEval, LLM-as-Judge) • Retrieval Metrics (NDCG, MRR, Precision/Recall@k) • Intent Classification • Model Serving & Quantization (vLLM, AWS Bedrock) • **Python:** PyTorch, Transformers, sentence-transformers, LangChain, LangGraph, LangSmith, NumPy, Pandas, scikit-learn

Cloud / On-Prem / DevOps:

Hybrid Cloud & On-Prem Infrastructure • AI Governance & Data Privacy • AWS (EC2, S3, Lambda, EKS, Bedrock) • Docker • Kubernetes • CI/CD Pipelines • GitHub Actions • Traefik • Observability/Tracing

CERTIFICATIONS

Anthropic — Claude Certified Architect: Foundational (CCA-F) | Issued June 5, 2026 · Valid through June 5, 2027 · Verify: credly.com/users/john-ryu

KEY PROJECTS

TecAce AI Agent Studio - Governance-First Enterprise Agentic Platform

Hybrid search (BM25 sparse + dense vectors), LangGraph agent workflows, guardrails + RAGAS/LLM-judge evaluation framework (FinanceBench-validated), and multi-platform chat integrations — deployable on hybrid cloud or fully on-prem infrastructure for data-sensitive teams. Tech: Python, FastAPI, LangChain/LangGraph, AWS Bedrock, on-prem vLLM, Milvus

MarketScopeAI (omnidora.com) - AI Marketing Automation SaaS

AI employees that qualify leads, nurture prospects, and run marketing playbooks 24/7 for small businesses. Tech: Next.js, TypeScript, AI Agents

PlinthPrep (plinthprep.com) - CCA-F Exam Prep Platform

Practice-exam platform for Anthropic's CCA-F certification with AI-generated question variations, cheat sheets, and decision trees. Tech: Next.js, Claude API

RecompIQ (recompiq.com) - Health Data Coaching App

Body-recomposition tracker for protocols, nutrition, biomarkers, and workouts with evidence-graded insights and projections. Tech: Next.js, TypeScript, Data Visualization

EDUCATION

Bachelor of Science in Computer Science

Temple University | College of Science & Technology | Philadelphia, PA